PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn

Taewon Song and Yeunwoong Kyung

***Abstract*—Modern Wi-Fi deployments suffer from spectrum underutilization. While an overlapping basic service set (OBSS) transmission occupies the primary channel, adjacent non-primary channels sit idle because conventional IEEE 802.11 mandates that the primary channel be included in every access. IEEE 802.11bn introduces non-primary channel access (NPCA) to exploit these idle intervals. The standard contention mechanism, however, is ill-suited to NPCA's constrained transmission windows, as its collision-triggered backoff growth synchronizes stations into repeated collisions or pushes their counters beyond the remaining opportunity, wasting airtime that could carry data. We propose PACE, a probabilistic adaptive contention control scheme that lets stations track the time-varying fair-share transmission probability of the depleting window without central coordination. PACE combines a deadline-scaled initialization and a distributed multiplicative feedback rule that copies the successful sender's probability on solo receptions, together with a prospective self-exclusion mechanism that silences stations whose frame length exceeds the remaining window. We evaluate PACE under IEEE 802.11 standard timing on a non-primary channel shared with native stations, against the standard-compliant NPCA baseline and a genie-aided fair-share reference. Across 5–50 visiting stations, PACE keeps the total useful airtime within 2% of the fair-share reference under RTS/CTS-protected access, up to 63% above the standard procedure and up to $4.5\times$ under basic access, while holding the visitors' airtime at or below their population-proportional share. The advantage grows with the visitor population, where standard backoff idles behind frozen counters and collides in synchronized bursts. These results establish probabilistic, deadline-aware contention as a design reference for non-primary channel access in next-generation Wi-Fi.**

***Index Terms*—Non-Primary Channel Access, IEEE 802.11bn, Probabilistic Access Control, OBSS, Adaptive Contention, Wi-Fi**

I. Introduction

Spectrum efficiency in next-generation Wi-Fi networks depends critically on how effectively stations (STAs) exploit available channel time. In dense deployments, overlapping basic service sets (OBSSs) routinely occupy the primary channel while adjacent channels remain idle, creating a persistent underutilization problem. Hence, IEEE 802.11bn addresses this gap by introducing non-primary channel access (NPCA), which allows STAs to transmit opportunistically on non-primary channels during ongoing OBSS intervals [1], [2].

NPCA, however, presents a contention challenge distinct from conventional channel access. Each opportunity lasts only as long as the ongoing OBSS transmission, so competing STAs must resolve their access within a finite, depleting window. To govern this access, the IEEE 802.11bn draft has STAs on the non-primary channel reuse the same binary exponential backoff (BEB) mechanism that the distributed coordination function (DCF) employs on the primary channel [1], yet BEB was designed for open-ended contention. On the primary channel, doubling the backoff window after each collision harmlessly spreads out retransmissions; within a bounded NPCA window the same growth turns harmful, as the inflated backoff frequently exceeds the time remaining. STAs whose counters overshoot the deadline are silenced for the rest of the interval, leaving the channel idle even when they still have data to send. Heterogeneous frame lengths compound the problem, since the draft leaves unspecified what a STA should do when its frame no longer fits the remaining window; a compliant STA may keep contending, or transmit only to be cut off at timer expiry, despite having no viable transmission opportunity.

Existing work on NPCA has focused on characterizing aggregate throughput and delay under the standard switching rule, taking BEB within the non-primary channel as given [2]–[4]. The contention dynamics within the finite NPCA window, and the mismatch between BEB's infinite-horizon design and the window's bounded duration, have not been addressed. Adaptive probabilistic schemes designed for infinite-horizon neighbor discovery, such as the ALOHA-like Neighbor Discovery (AND) algorithm [5], reduce overhead by avoiding explicit coordination, but their stationary-population assumption makes them ill-suited to a window that depletes over time. BEB's infinite-horizon design and AND's open-loop phase structure both fail to track the time-varying optimal access probability as the window shrinks and the set of viable STAs changes.

In this paper, we propose PACE, a probabilistic adaptive contention control scheme for NPCA that enables STAs to collectively track the time-varying fair-share transmission probability of each moment within the window, without central coordination or explicit inter-STA signaling. PACE adapts the MIMD feedback rule inspired by the probabilistic neighbor discovery (PND) algorithm [6] to the finite-window NPCA

T. Song is with the Division of Electronics and Electrical Engineering, College of Engineering, Dongguk University, 30, Pildong-ro 1-gil, Jung-gu, Seoul 04620, Republic of Korea (e-mail: twsong@dongguk.edu) and Y. Kyung is with the Department of Electronic Engineering, Seoul National University of Science and Technology, Korea. (e-mail: ywkyung@seoultech.ac.kr), Corresponding author: Y. Kyung

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TABLE I: Comparison of Contention Control Schemes

Method	Mechanism	Objective	PPDU-aware	Signaling-free	NPCA target
AND [5]	Fixed-probability slotted ALOHA	Discover all neighbors in finite time without coordination	×	✓	×
PND [6]	Multiplicative feedback with solo-copy rule	Converge transmission probability to optimal without the contender count	×	✓	×
Cali *et al.* [7]	p-persistent with contender estimation	Approach theoretical throughput limit in a distributed manner	×	✓	×
BLADE [8]	Hybrid increase / multiplicative decrease via access rate	Eliminate packet-delivery droughts for real-time Wi-Fi	×	✓	×
IYT [9]	Neighbor-ordered backoff interval assignment	Reduce collision probability and worst-case latency	×	✓	×
ACWB-DQN [10]	Deep Q-network contention window selection	Maximize throughput under varying network load	×	×	×
FC-MAPPO [11]	Fairness-constrained multi-agent policy optimization	Minimize collisions while ensuring inter-STA fairness	×	×	×
PACE (proposed)	**Multiplicative feedback with PPDU-aware self-exclusion**	**Maximize channel utilization in finite NPCA window**	✓	✓	✓

context. STAs copy the successful sender's probability on solo receptions, reduce on collision, and increase on idle slots, causing distributed dynamics to track the fair-share policy. PACE further adds a PPDU-aware self-exclusion rule that silences STAs whose frame length exceeds the remaining window, eliminating a class of futile contention attempts that BEB would otherwise sustain.

The design of PACE is motivated by two requirements that are unique to NPCA's bounded-window structure: 1) the fair-share transmission probability is time-varying and must track the depleting window rather than converge to a static equilibrium, and 2) STAs whose frame length exceeds the remaining window must proactively self-exclude to prevent futile contention that wastes viable slots. The main contributions of this paper are as follows.

- We characterize the fair-share transmission probability for finite-window NPCA, showing that it must continuously track the remaining window duration and the number of still-viable competing STAs, a time-varying, rising target that neither BEB nor existing infinite-horizon adaptive schemes can follow.
- We propose PACE, a fully distributed algorithm that addresses both requirements simultaneously. A deadline-scaled initialization and MIMD feedback drive the transmission probability toward the time-varying target without inter-STA signaling or knowledge of the total STA count.
- Through simulation in IEEE 802.11 standard timing on a channel shared with native DCF stations, we demonstrate that PACE keeps the total useful airtime within 2% of a genie-aided fair-share reference under RTS/CTS-protected access and up to $4.5\times$ above the standard-compliant baseline under basic access, holds the visitors' airtime at or below their population-proportional share, and attains a three-fold higher per-attempt success ratio as shown in the ablation study.

The remainder of this paper is organized as follows. Section II reviews related work on NPCA and adaptive MAC protocols. Section III presents the system model. Section IV describes the PACE algorithm and its theoretical foundation. Section V presents simulation results, and Section VI concludes the paper.

II. Related Work

Table I summarizes the key distinctions among existing contention control schemes and the proposed PACE.

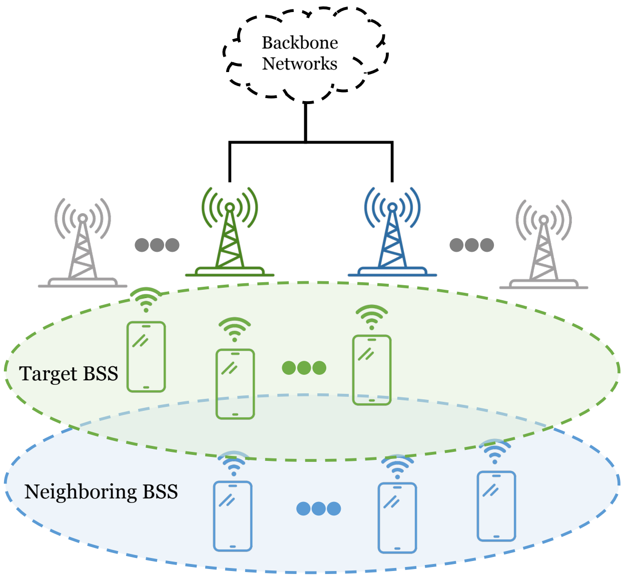
A. Non-Primary Channel Access in IEEE 802.11bn

IEEE 802.11bn introduces NPCA as a core mechanism for exploiting idle secondary channels during OBSS transmissions [1], [12]. Wei *et al.* [2] extended Bianchi's DCF model to a multi-channel setting, establishing analytically that NPCA always outperforms the non-NPCA mode; a follow-up study [4] incorporated switching overhead and proposed a threshold policy to toggle between NPCA and legacy modes based on primary channel occupancy. Bellalta *et al.* [3] developed a Continuous-Time Markov Chain model for a multi-BSS topology, identifying a secondary-channel contention trade-off that can limit NPCA gains when the secondary channel is shared. Lee and Song [13] further showed that PPDU frame duration has a nonlinear effect on inter-BSS coexistence, and Song [14] applied deep reinforcement learning to the NPCA channel-activation decision.

Across these studies, however, efforts to improve NPCA performance target the decision of *when* to switch to the non-primary channel, through occupancy-based threshold policies [4] or learned activation rules [14], while the contention mechanism exercised *within* the window is left fixed at BEB. How STAs should adapt their access probability as the window depletes and viable STAs drop out has not been studied.

B. Adaptive Contention Control in Wireless LANs

Bianchi [15] established that BEB operates below the theoretical throughput ceiling, and Cali *et al.* [7] showed that p-persistent access with dynamic contender estimation can approach this limit in a distributed manner. Subsequent work has diversified the adaptation mechanisms. BLADE [8] uses a HIMD policy driven by a universally observable channel



	Channel #1	Channel #2
Target BSS	BSS Primary	NPCA Primary
Neighboring BSS		BSS Primary
Overlapping BSS	OBSS Tx	

Fig. 1: Two-BSS topology and channel assignment.

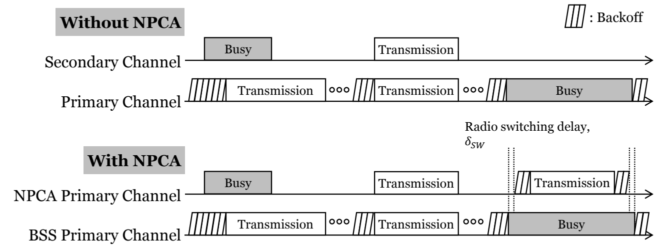


Fig. 2: The medium access process comparison with and without NPCA.

signal; Wilhelmi *et al.* [9] impose structured transmission order via inter-BSS activity awareness; and learning-based approaches optimize the contention window through deep Q-networks [10] or fairness-constrained multi-agent policy optimization [11].

Despite this breadth, all existing schemes assume STAs contend over an open-ended horizon and target steady-state convergence. NPCA's bounded window demands a fundamentally different approach. The optimal transmission probability must evolve as the window shrinks and the viable STA population changes. The MIMD feedback rule of PND [6] is the closest mechanism in spirit, but was designed for infinite-horizon neighbor discovery and provides no accounting for bounded window duration or frame-length viability.

III. System Model

A. Network and Channel Model

Fig. 1 depicts the two-BSS topology and channel assignment assumed throughout this work. The target BSS, whose NPCA operation is our focus, comprises an access point (AP) and a set $\mathcal{N} = \{1, 2, \ldots, N_{\text{vis}}\}$ of associated non-AP STAs and they can be considered as *visitors* to the neighboring BSS. The visitors normally contend on their BSS primary channel, Channel #1 in Fig. 1, which is the main channel used by the target BSS for its own transmissions.

Channel #1 is also used by an OBSS operating on the same channel, whose transmissions intermittently render it busy. In standards prior to IEEE 802.11bn, a STA could not initiate a transmission whenever its designated primary channel was busy, even if the secondary channel was idle. In IEEE 802.11bn, however, the AP can designate a separate channel as the NPCA primary channel, used by the visitors during NPCA operation, named as Channel #2 in Fig. 1. We use the qualifier "BSS" to distinguish the original primary channel from the NPCA primary channel, consistent with P802.11bn Draft terminology [1].

The NPCA primary channel is not unused spectrum reserved for the target BSS. Channel #2 is itself the BSS primary channel of the neighboring BSS, comprising its own AP and a set $\mathcal{M} = \{1, 2, \ldots, N_{\text{nat}}\}$ of *native* STAs that contend on it with standard DCF at all times. The visitors, by contrast, use Channel #2 only opportunistically, during NPCA visits. Two classes of contenders therefore coexist on the NPCA primary channel: the native STAs, which contend there persistently, and the visitors, whose access is confined to the bounded NPCA window.

As depicted in Fig. 1, we assume the coverages of the two BSSs coincide, so that all visitors and native STAs are mutually within carrier-sensing range and form a single collision domain on the NPCA primary channel: every visitor senses, defers to, and competes with every native STA. This is the configuration in which the visitors' impact on native airtime is greatest, so the coexistence results reported here upper-bound that cost. In general the two coverages may overlap only partially, in which case a visitor outside the neighboring BSS coverage would reuse the NPCA primary channel without contending with the natives at all; this spatial-reuse regime only lowers the coexistence cost and is left to future work.

B. NPCA Operation and Effective Window

Fig. 2 contrasts the medium-access processes with and without NPCA operation. Under conventional access (top), prior to IEEE 802.11bn, a STA contends only on the BSS primary channel.[1] It decrements its backoff counter while the channel is idle and transmits when the counter reaches zero. If the secondary channel is also sensed idle at this instant, the STA may bond it with the BSS primary channel and transmit a wideband PPDU spanning the combined bandwidth (e.g., 40,

[1] We use the term BSS primary channel to distinguish the original primary channel from the NPCA primary channel introduced in IEEE 802.11bn. Prior to 802.11bn, when NPCA was not yet defined, this channel was referred to simply as the primary channel.

80, 160, or 320 MHz). When an OBSS transmission seizes the BSS primary channel, however, the STA must defer and freeze its backoff for the entire OBSS busy period, even though the secondary channel is idle.

Under NPCA access (bottom), a STA no longer wastes the OBSS busy period. When it detects an OBSS transmission on the BSS primary channel, it decodes the preamble of that transmission and learns its duration in advance,[2] rather than freezing its backoff for the whole period, it switches to the idle NPCA primary channel and contends and transmits there. On each such switch the STA saves its backoff state on the BSS primary channel and initializes a fresh contention window for the NPCA primary channel, so every NPCA visit begins at the minimum backoff stage regardless of prior history. The STA already knows when the OBSS transmission will finish, so it sets a timer to that instant and switches back to the BSS primary channel when the timer expires, restoring the saved state without having to re-sense the channel. In this way the secondary spectrum that the a conventional STA leaves idle is put to use, and a STA can complete one or more transmissions that it would otherwise have had to postpone.

Two constraints shape this access. First, a radio transition delay $\delta_{\rm sw}$ is incurred on each switch: the STA becomes ready on the NPCA primary channel only after the switching delay, and the NPCA timer is set to expire one switch-back delay before the OBSS transmission ends so that the STA is back on the BSS primary channel in time [1]. Second, the opportunity lasts only until the known end of the OBSS transmission, whose remaining duration $D_{\rm rem}$ is read from the preamble. Contention and transmission must therefore fit within a bounded window of

$$W_{\rm eff} \;=\; \left\lfloor \frac{D_{\rm rem} - 2\,\delta_{\rm sw}}{\sigma} \right\rfloor \quad \text{slots}, \tag{1}$$

where σ is the slot time, typically 9 μs in IEEE 802.11-compliance systems. This deadline is the crux of NPCA. Every backoff decision competes against it. A STA that collides doubles its contention window as usual, and once the resulting backoff exceeds the slots left in $W_{\rm eff}$, it can no longer transmit before the window closes so it wastes the rest of the opportunity despite having data to send.

Because the contention window is reset at every NPCA visit, this penalty recurs each time rather than amortizing across visits. This mismatch between open-ended backoff and the finite window is the inefficiency that PACE targets.

C. Slotted Contention Model

We model time on the NPCA primary channel as a sequence of slots of duration σ. Each visitor maintains a per-slot transmission probability τ_i and, when eligible, transmits in a slot with that probability, in the manner of slotted ALOHA.

[2]NPCA relies on preamble detection rather than energy detection. A STA must decode the inter-BSS PPDU preamble (L-SIG and BSS color) at PHY-RXSTART to both classify the transmission and derive its remaining duration, from which the NPCA window is determined. Energy detection alone, which yields no length or BSS information, is insufficient to trigger an NPCA transition.

This per-slot access probability is the variable that the scheme of Section IV controls. The standard baseline instead runs its backoff countdown unchanged, and enters the model only through the slot outcomes defined below.

This probabilistic access departs from the current draft, which reuses the EDCA backoff procedure on the NPCA primary channel [1]. PACE keeps the NPCA architecture intact, including the switching triggers, the NPCA timer, and the contention-state save and restore, and replaces only the contention rule applied within the bounded visit.

We index the slots by $t \in \{0, 1, \ldots, W_{\rm eff}\}$. We denote by $W_{\rm rem}(t) = W_{\rm eff} - t$ the number of slots remaining at slot t, so that $W_{\rm rem}$ counts down toward the deadline as contention proceeds. Let $\mathcal{V} \subseteq \mathcal{N}$ be the set of STAs that switch to the NPCA primary channel, each holding a pending PPDU whose transmission occupies L_i slots. Because a transmission must complete before the window closes, STA i is viable at slot t only if its frame still fits, i.e. $L_i \leq W_{\rm rem}(t)$; otherwise it can no longer succeed within the visit. We write the viable population at slot t as

$$\mathcal{V}(t) \;=\; \left\{\, i \in \mathcal{V} : L_i \leq W_{\rm rem}(t) \,\right\}, \tag{2}$$

which is non-increasing in t as both the window depletes and frames become infeasible.

At each slot, a viable STA i attempts transmission with probability τ_i. Following the half-duplex assumption, a transmitting STA cannot sense the channel in the same slot. Let $\mathcal{D}(t)$ denote the set of STAs whose transmissions on the NPCA primary channel are detected by a given STA in slot t. This set is defined at the channel level. It counts every STA transmitting on that channel in the slot, not only the switching visitors but also any other STA that uses the same channel as its own primary. The slot outcome is idle when $|\mathcal{D}(t)| = 0$, a success when $|\mathcal{D}(t)| = 1$, and a collision when $|\mathcal{D}(t)| \geq 2$. These three observable events are the only feedback available to a STA, and they drive the adaptive contention control developed in Section IV.

The remaining window advances by the time each outcome occupies the medium. An idle slot consumes one slot, while a successful slot consumes $L_{\rm solo}(t)$ slots and a collision consumes $L_{\rm col}(t)$ slots. The two occupancies depend on the access mode.

1) Basic Access: A successful slot is occupied by the sole sender's frame,

$$L_{\rm solo}(t) \;=\; L_i \tag{3}$$

where $\mathcal{D}(t) = \{i\}$, while a collision keeps the medium busy until the longest colliding frame has ended,

$$L_{\rm col}(t) \;=\; \max_{i \in \mathcal{D}(t)} L_i, \tag{4}$$

as in standard IEEE 802.11 without collision detection. Since each STA holds a single pending PPDU per visit and departs upon success, every L_i is fixed for the visit's duration; the slot dependence of both occupancies enters only through the identity of the transmitters $\mathcal{D}(t)$.

2) RTS/CTS-Protected Access: Under RTS/CTS-protected access the busy-slot accounting changes. A successful exchange is preceded by the handshake and therefore occupies

$$L_{\mathrm{solo}}(t) \;=\; L_i + L_{\mathrm{hs}} \tag{5}$$

slots, where $\mathcal{D}(t) = \{i\}$ and

$$L_{\mathrm{hs}} \;=\; \left\lceil \frac{T_{\mathrm{RTS}} + T_{\mathrm{CTS}} + 2\,T_{\mathrm{SIFS}}}{\sigma} \right\rceil \tag{6}$$

is the handshake overhead, with T_{RTS}, T_{CTS}, and T_{SIFS} being the respective durations defined in IEEE 802.11. A collision, by contrast, involves only RTS frames. The data frames are never sent, and the channel recovers after the extended interframe space, so a collision occupies

$$L_{\mathrm{col}} \;=\; \left\lceil \frac{T_{\mathrm{RTS}} + T_{\mathrm{EIFS}}}{\sigma} \right\rceil \tag{7}$$

slots, where $T_{\mathrm{EIFS}} = T_{\mathrm{SIFS}} + T_{\mathrm{ACK}} + T_{\mathrm{DIFS}}$ is the extended interframe space. Unlike its basic-access counterpart of (4), this occupancy is a constant, independent of the colliding frame lengths and of the slot index. Under protection the viability test of (2) must also leave room for the handshake, i.e. L_i is replaced by $L_i + L_{\mathrm{hs}}$. Basic access is recovered as the special case $L_{\mathrm{hs}} = 0$ with $L_{\mathrm{col}}(t)$ given by (4). Both access modes are evaluated in Section V-A.

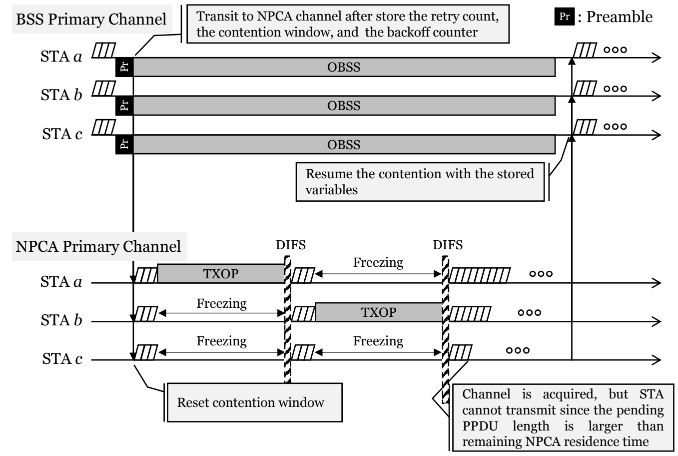
IV. PACE Algorithm

Fig. 3 illustrates contention on the NPCA primary channel under the slotted model of Section III-C and contrasts the two schemes studied in this paper. In both, a STA that detects an OBSS transmission on the BSS primary channel stores its DCF state, migrates to the NPCA primary channel, and contends there until the NPCA timer expires, whereupon it restores the saved state and resumes on the BSS primary channel. Under the NPCA baseline of Fig. 3a, STAs contend with BEB. A STA that collides doubles its contention window, so its counter can outlast the remaining window, and because the countdown is frozen during every busy period, the surviving counters tend to expire together and collide again.

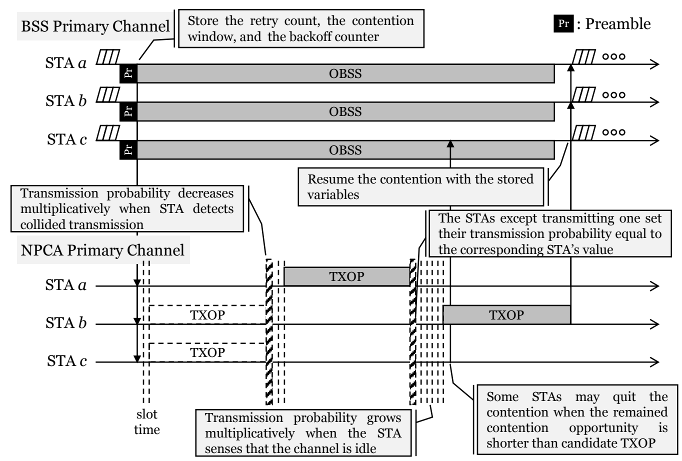
Meanwhile, under the proposed PACE scheme of Fig. 3b, each STA instead adapts its per-slot probability τ_i from the idle, success, and collision feedback. The access probability is raised multiplicatively on idle slots, lowered on collisions, and replaced by the successful sender's value on solo receptions. In addition, a STA proactively withdraws from contention once its frame can no longer fit within $W_{\mathrm{rem}}(t)$.

Algorithm 1 summarizes PACE as executed independently by each STA on the NPCA primary channel during a single NPCA visit. Every STA runs the same rule using only locally observable feedback, those are, the per-slot outcome and its own remaining window without central coordination or knowledge of the contending population.

Algorithm 1 is expressed in the per-slot transmission-probability model of Section III-C. A viable STA transmits in a slot with probability τ_i. This is a probabilistic, slotted-ALOHA-style access policy.



(a) NPCA baseline (BEB).



(b) Proposed PACE (probabilistic adaptive contention).

Fig. 3: Medium access on the NPCA primary channel: a STA that detects an OBSS transmission on the BSS primary channel migrates and contends within the bounded NPCA window. (a) the standard NPCA baseline with binary exponential backoff, and (b) the proposed PACE with probabilistic adaptive contention and PPDU-aware self-exclusion.

We adopt the probabilistic approach for two reasons. First, the per-slot probability τ is the common abstraction under which DCF, PND, and AND are all analyzed [5], [6], [15], so it places PACE on the same analytical footing as these schemes. Second, and more fundamentally, the finite NPCA window is precisely where a deterministic countdown breaks down. A single backoff draw cannot re-adapt within the handful of slots a visit lasts, whereas a per-slot probability can be re-steered every slot by the feedback law, which will be described in Subsec. IV-D. The ALOHA-style attempt is therefore not incidental but the mechanism that makes within-window adaptation possible.

Algorithm 1 runs four phases per visit. (a) a one-time probability initialization, followed by a per-slot loop of (b) self-exclusion, (c) channel access, and (d) feedback-based

Algorithm 1 PACE at STA i during one NPCA visit

Require: PPDU length L_i (slots), window W_{eff}, factors $c_{\text{coll}}, c_{\text{idle}} > 1$, handshake overhead L_{hs} of Eq. (6) ($L_{\text{hs}} = 0$ under basic access)

/* Phase (a): Probability Initialization */
$\tau_i \leftarrow 1/W_{\text{eff}}$; $\quad W_{\text{rem}} \leftarrow W_{\text{eff}}$
while $W_{\text{rem}} > 0$ **do**
  /* Phase (b): PPDU-aware Self-Exclusion */
  if $L_i + L_{\text{hs}} > W_{\text{rem}}$ **then**
    $\tau_i \leftarrow 0$
  end if
  /* Phase (c): Probabilistic Channel Access */
  draw $u \sim \text{Uniform}(0, 1)$
  if $u < \tau_i$ **then**
    transmit the PPDU, advertising τ_i (preceded by the RTS/CTS handshake under protected access)
    if solo success **then**
      break
    end if
    $W_{\text{rem}} \leftarrow W_{\text{rem}} - L_{\text{col}}(t)$ {collided: Eq. (7) under protection, Eq. (4) under basic access}
  else
    /* Phase (d): Feedback-based Adaptation */
    observe the slot outcome $|\mathcal{D}(t)|$
    if $|\mathcal{D}(t)| = 0$ **then**
      $\tau_i \leftarrow \min(\tau_i \cdot c_{\text{idle}}, 1)$; $\quad W_{\text{rem}} \leftarrow W_{\text{rem}} - 1$
    else if $|\mathcal{D}(t)| = 1$ **then**
      $\tau_i \leftarrow \tau_{\text{sender}}$; $\quad W_{\text{rem}} \leftarrow W_{\text{rem}} - L_{\text{solo}}(t)$
    else
      $\tau_i \leftarrow \tau_i / c_{\text{coll}}$; $\quad W_{\text{rem}} \leftarrow W_{\text{rem}} - L_{\text{col}}(t)$
    end if
  end if
end while

adaptation. The remainder of this section develops each phase in turn.

A. Phase (a): Probability Initialization

A visitor that has just switched to the NPCA primary channel does not know how many other STAs are contending there. The population-matched choice $\tau_0 = 1/(N_{\text{vis}}+N_{\text{nat}})$, which is the well-known throughput-optimal attempt probability of slotted ALOHA, is therefore unavailable in practice since the contender count is not observable.

Visitor STAs might, through association state or management traffic, form some estimate of how many STAs belong to its own BSS. The harder obstacle is that the population it actually contends with also includes the native STAs of the neighboring BSS that owns the NPCA primary channel. That is, a visitor competes against these as well, yet has no direct way to learn how many are active. The count that would parameterize $1/(N_{\text{vis}}+N_{\text{nat}})$ is thus doubly hidden, which motivates a surrogate that avoids counting STAs altogether.

We propose a method where the window length itself determines the scale. A visit spans W_{eff} slots, and a STA contends in at most one of them per slot so W_{eff} upper-bounds the number of attempt opportunities any STA sees in a visit. A STA attempting with probability τ therefore expects at most $W_{\text{eff}}\,\tau$ attempts before the deadline. Fixing this budget to a single probe per STA,

$$W_{\text{eff}}\,\tau_0 = 1, \tag{8}$$

sets the most conservative self-consistent exploration rate. Low enough to avoid an early collision storm, yet high enough for each STA to test the medium about once before the window drains. This yields

$$\tau_0 = \frac{1}{W_{\text{eff}}}, \tag{9}$$

which depends only on the effective window W_{eff}, which is a quantity that each STA can compute locally from the OBSS PPDU duration. No knowledge of the contender count is required.

In general frame length to be transmitted on NPCA primacy channel is longer than a single slot, so τ_0 is bound to be smaller than the optimal value. This choice is deliberately biased toward under-attempting for two reasons. First, it is self-correcting in the safe direction. An over-conservative τ_i is raised toward the operating point within a few slots by the idle-driven multiplicative increase in Phase (d), whereas an over-aggressive start triggers collisions whose wasted slots cannot be reclaimed inside a finite window. The MIMD recovery is thus asymmetric, and starting low pays only a bounded ramp-up cost. Second, $1/W_{\text{eff}}$ tracks the regime where initialization matters most. When the window is tight ($W_{\text{eff}} \approx N_{\text{vis}}+N_{\text{nat}}$) it already approximates the optimum $1/(N_{\text{vis}}+N_{\text{nat}})$, and when the window is loose ($W_{\text{eff}} \gg N_{\text{vis}}+N_{\text{nat}}$) the mild initial timidity is erased by the idle increase long before the window drains.

B. Phase (b): PPDU-aware Self-Exclusion

At the start of every slot a STA compares its own frame length L_i to the remaining window $W_{\text{rem}}(t)$. If $L_i > W_{\text{rem}}(t)$ the frame can no longer finish before the deadline, so the STA sets $\tau_i = 0$ and falls silent for the rest of the visit. The test is purely local. That means it involves only the STA's own L_i and the slot index and thus, like initialization, requires no knowledge of the contending population.

Self-exclusion keeps the optimal target that PACE aims to track,

$$\tau^*(t) \;=\; \frac{1}{|\mathcal{V}(t)|}, \tag{10}$$

meaningful in a heterogeneous and finite window. In a slot with $|\mathcal{V}(t)|$ viable STAs, a success, meaning exactly one transmitter, is most likely when each attempts with probability $\tau^*(t)$, and this target rises over the visit as the viable set drains. A STA whose frame no longer fits is removed from the viable set $\mathcal{V}(t)$ of Eq. (2), so it neither wastes time on attempts destined to fail nor inflates the contention seen by the STAs that can still succeed. This plays the same role as the collision detection departure of the original neighbor discovery algorithm, in which a STA leaves the contention once its advertisement succeeds. Here a STA also leaves once its frame provably cannot complete, tightening $\mathcal{V}(t)$ toward the set of STAs that genuinely compete for the remaining slots.

C. Phase (c): Probabilistic Channel Access

A STA that is still viable transmits in the current slot with probability τ_i and otherwise listens. Because the radio is half-duplex, a transmitting STA cannot sense the slot it occupies, so it performs no adaptation on its own transmissions. Each transmitted PPDU advertises the sender's current τ_i in its header, which the listeners use in Phase (d).

A slot is a success when exactly one viable STA transmits and its frame fits the remaining window. The winner then departs the visit, mirroring the post-success departure of neighbor discovery, which decrements $|\mathcal{V}(t)|$ and raises the optimal target $\tau^*(t)$ for the STAs that remain. Slots with no transmission are idle, and slots with two or more transmissions are collisions; both outcomes are observed by every non-transmitting STA and feed the adaptation of Phase (d).

D. Phase (d): Feedback-based Adaptation

Every viable STA that listened in the slot updates τ_i from the slot outcome $|\mathcal{D}(t)|$ using the multiplicative-increase and multiplicative-decrease (MIMD) rule inherited from neighbor discovery as

$$\tau_i \leftarrow \begin{cases} \min(c_{\mathrm{idle}}\,\tau_i,\,1) & |\mathcal{D}(t)| = 0 \quad \text{(idle)}, \\ \tau_{\mathrm{sender}} & |\mathcal{D}(t)| = 1 \quad \text{(solo success)}, \\ \tau_i / c_{\mathrm{coll}} & |\mathcal{D}(t)| \geq 2 \quad \text{(collision)}, \end{cases} \tag{11}$$

with $c_{\mathrm{idle}}, c_{\mathrm{coll}} > 1$.

An idle slot signals that the population has been overestimated and the attempt rate is raised by the factor of c_{idle}. On the other hand, a collision signals the opposite, hence, once STAs detect a collision, the attempt rate is divided by c_{coll} to reduce the contention. On a solo success, every listener copies the advertised τ_{sender}, so a single successful slot propagates the winning operating point to the whole contending set at once, giving PACE the fast intra-visit consensus.

None of the three updates counts the viable STAs. Instead, the idle and collision statistics move each τ_i toward the operating point implicitly, and solo-copy pulls the population onto a common value, so the ensemble approximately tracks the time-varying optimum $\tau^*(t)$ of Eq. (10) without estimating the number of contending STAs $|\mathcal{V}(t)|$.

The direction of this adaptation is what separates PACE from BEB. Because the viable set only shrinks over a visit, the optimal target $\tau^*(t) = 1/|\mathcal{V}(t)|$ of (10) rises as the deadline approaches. On the other hand, the BEB moves the opposite way, that is, a collision doubles the contention window and halves the attempt rate, so the access probability falls exactly when it should climb.

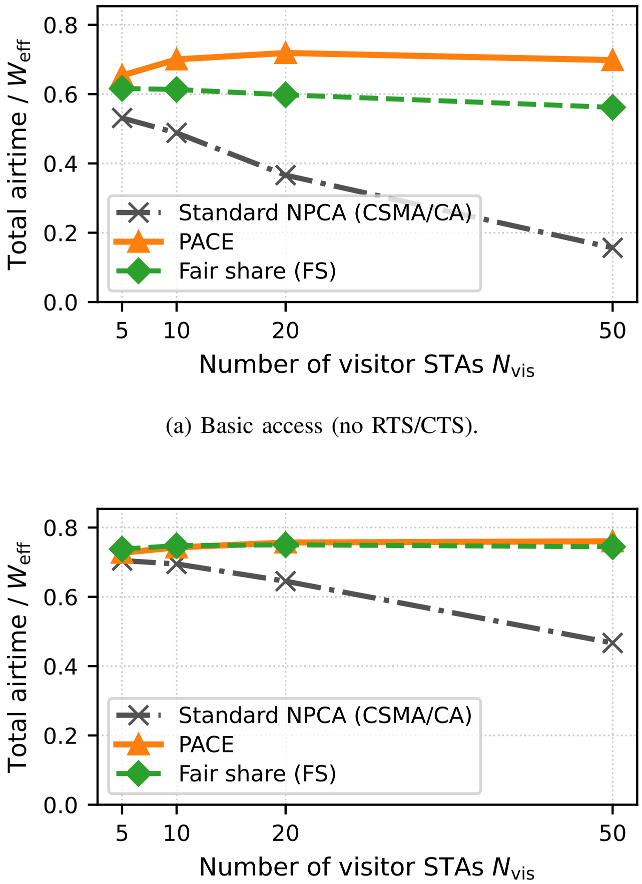
Finally, the rule of Eq. (11) needs no modification to operate alongside the native STAs of Section III-A. Because the slot outcome $\mathcal{D}(t)$ is measured at the channel level, native activity enters a visitor's feedback directly. A slot in which two or more frames overlap is a collision whether the extra transmitter is a visitor or a native, so a native transmission that clashes with a visitor lowers that visitor's τ_i, and a slot with no transmission at all, native or visitor, is idle and raises it. The solo case is the one exception. The copy $\tau_i \leftarrow \tau_{\mathrm{sender}}$ requires the successful frame to advertise a probability, which only a PACE visitor does; when a native transmits alone, its frame carries no such value, so a listening visitor keeps τ_i unchanged and simply observes the busy slot while the window continues to shrink.

TABLE II: Simulation Parameters

Parameter	Value
Visitor STAs N_{vis} (scheme under test)	$\{5, 10, 20, 50\}$
Native STAs N_{nat} (always standard DCF)	10
Slot time σ / SIFS / DIFS	9 / 16 / 34 μs
Effective window W_{eff}	420 slots (3.78 ms)
Visitor PPDU duration	$\mathcal{U}[225, 900]$ μs
Native PPDU duration	450 μs
Initial probability τ_0	$1/W_{\mathrm{eff}}$
MIMD factors $c_{\mathrm{coll}}, c_{\mathrm{idle}}$	1.2
$\mathrm{CW}_{\min}$ / $\mathrm{CW}_{\max}$ for BEB	15 / 1023
RTS/CTS L_{hs} / L_{col} (24 Mbps ctrl.)	88 / 106 μs
Transitions $\times$ sequences $\times$ seeds	$50 \times 20 \times 5$

V. Performance Evaluation

We evaluate PACE in the slotted NPCA contention model of Section III-C under IEEE 802.11 OFDM timing; Table II summarizes the parameters. A visit spans $W_{\mathrm{eff}} = 3.78$ ms (420 slots of $\sigma = 9$ μs), contended by $N_{\mathrm{vis}} \in \{5, 10, 20, 50\}$ visitor STAs running the scheme under test alongside $N_{\mathrm{nat}} = 10$ native STAs that always run standard DCF. Visitor PPDU durations are drawn uniformly from 225–900 μs and native frames last 450 μs. PACE follows Algorithm 1 with $\tau_0 = 1/W_{\mathrm{eff}}$ and $c_{\mathrm{coll}} = c_{\mathrm{idle}} = 1.2$. The standard-compliant baseline contends with CSMA/CA ($\mathrm{CW}_{\min} = 16$, BEB) as Draft 1.2 specifies it [1]. The NPCA Minimum Duration Threshold gates only the decision to switch, and the draft leaves unspecified what a visitor does when its pending frame no longer fits the remaining window. We therefore grant the baseline the stronger of the two conformant behaviors, a *deferring* implementation that checks the fit before transmitting and withholds unfittable frames, the same locally computable test that PACE's Phase (b) applies. The alternative conformant reading, transmitting anyway and being cut off at NPCA-timer expiry, performs strictly worse (e.g., total useful airtime 0.12 versus 0.16 at $N_{\mathrm{vis}} = 50$ under basic access, and 0.38 versus 0.63 for 0.9-ms visits under RTS/CTS), so adopting the deferring baseline is the conservative choice for our comparison. A genie-aided *fair-share* reference (FS) transmits at $\tau^*(t) = 1/|\mathcal{V}(t)|$ with perfect knowledge of the viable set; FS is not an airtime optimum but the attempt-fair operating point that PACE is designed to track. Native STAs, which neither know nor observe the visitors' window, likewise contend without any fit restriction; a native frame that straddles the window end completes on the channel, and its in-window portion is credited to the window utilization. Both access modes of Section III-C are evaluated. basic access, in which a collision is charged the longest colliding frame,



(a) Basic access (no RTS/CTS).

(b) Mandatory RTS/CTS.

Fig. 4: Total useful airtime (visitor and native combined) versus the number of visitor STAs N_{vis} for standard-compliant NPCA (CSMA/CA visitors), PACE, and the fair-share reference (FS), in the mixed native/visitor channel with $N_{\text{nat}} = 10$ native DCF stations, under IEEE 802.11 timing ($W_{\text{eff}} = 3.78$ ms, visitor PPDU 225–900 μs, native PPDU 450 μs).

eq. (4), and mandatory RTS/CTS with the standard-derived costs $L_{\text{hs}} = 88$ μs and $L_{\text{col}} = 106$ μs. Each data point averages the steady half of 50 consecutive NPCA transitions over 20 sequences and five seeds. Performance is reported as the total useful airtime of the channel normalized by W_{eff}, its split between visitors and natives, and the visitor airtime proportionality index defined in Section V-B.

A. Why PACE: Channel Airtime under Mixed Contention

We first ask the question that motivates this work. What does PACE buy over simply running the standard access procedure on the non-primary channel? The design objective of PACE is not visitor throughput per se, but to keep the shared NPCA channel *efficient and fair*. The FS policy $\tau^*(t) = 1/|\mathcal{V}(t)|$, computed over *all* contenders, is precisely the embodiment of that objective, and the evaluation therefore reports the *total* useful airtime of the channel. Fig. 4 compares $N_{\text{vis}} \in \{5, \ldots, 50\}$ visitor STAs operating either standard CSMA/CA ($\text{CW}_{\min} = 16$, binary exponential backoff, with the NPCA minimum-duration check) or PACE exactly as specified in Algorithm 1, every visit starts cold from $\tau_0 = 1/W_{\text{eff}}$, sharing the channel with $N_{\text{nat}} = 10$ native DCF stations, using IEEE 802.11 OFDM timing ($\sigma = 9$ μs).

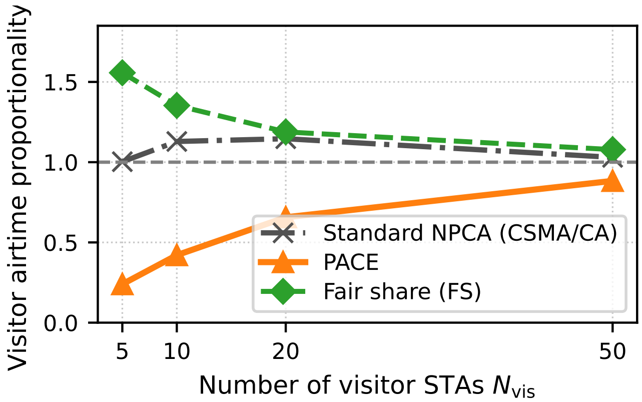
Under basic access (Fig. 4a), the channel under standard-compliant visitors degrades monotonically as the visitor population grows, from 0.53 at $N_{\text{vis}} = 5$ to 0.16 at $N_{\text{vis}} = 50$. The fixed $\text{CW}_{\min} = 16$ is far below the contention level, and the idle-gated countdown synchronizes the visitors into collision cascades, each charged the full length of the longest colliding frame. The standard procedure thus does not merely starve the visitors; it destroys the channel. PACE instead holds the total airtime essentially flat at 0.65–0.72 across the entire range, above FS at every point, reaching $4.5\times$ the standard procedure at $N_{\text{vis}} = 50$. The composition behind this flat total is the fair-sharing behavior by design. At $N_{\text{vis}} = 5$ the deliberately timid start $\tau_0 = 1/W_{\text{eff}}$ leaves the channel largely to the natives (visitor share 0.05), and as the visitor population, and hence its legitimate demand, grows, the idle-driven increase raises the visitors' share monotonically to 0.51 at $N_{\text{vis}} = 50$, while self-exclusion returns slots that can no longer carry a full frame. PACE exceeds FS in total airtime here because FS enforces the per-STA fair attempt rate at the cost of collisions that basic access charges at full frame length, whereas PACE's conservative ramp avoids them.

Under RTS/CTS-protected access (Fig. 4b), a collision costs the constant L_{col} of Eq. (7) (106 μs) instead of a full frame, at the price of the L_{hs} handshake (88 μs) on every success. Collisions being cheap, the fair-share policy is now also airtime-efficient, and PACE tracks it within 2% over the whole range (0.73–0.76), while the channel under standard visitors declines from 0.71 to 0.47. The handshake caps what a collision costs, but neither the frozen-backoff idling nor the synchronized collision bursts are curable by RTS/CTS, leaving PACE with 63% more channel airtime at $N_{\text{vis}} = 50$. Taken together, the two panels justify PACE's premise. Inside a finite NPCA window shared with natives, probabilistic deadline-aware access keeps the channel near its efficient-and-fair operating point where the standard backoff idles or collides it away.

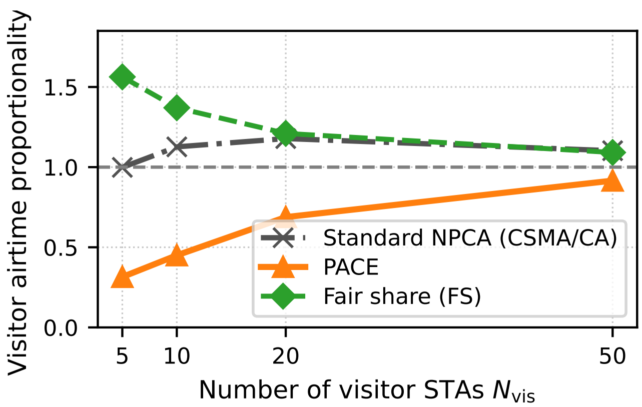
B. Fairness between Natives and Visitors

Total airtime alone does not reveal who receives it, so we next examine how the window is divided between the two populations. Fig. 5 reports the visitor airtime proportionality index ρ, the visitors' share of the useful airtime normalized by their population share $N_{\text{vis}}/(N_{\text{vis}} + N_{\text{nat}})$, in the identical setting as Fig. 4. $\rho = 1$ is proportional fairness, and $\rho > 1$ indicates that the visitors occupy the channel beyond their share.

FS itself sits well above unity ($\rho = 1.6$ at $N_{\text{vis}} = 5$, falling to 1.1 at $N_{\text{vis}} = 50$), which calibrates the reference. Even a per-STA fair attempt rate yields a visitor-leaning split, both because the visitor frames are longer than the native ones (562 μs versus 450 μs on average) and because the

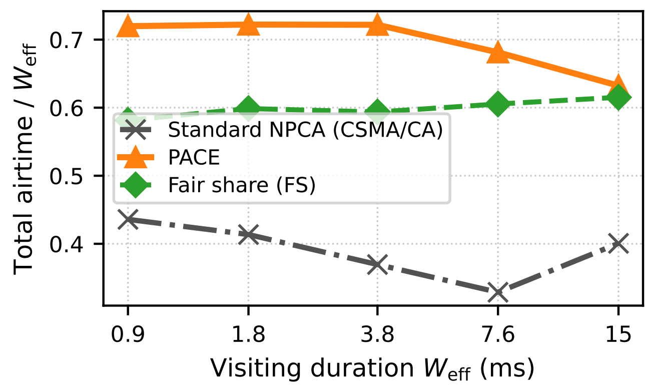


(a) Basic access (no RTS/CTS).

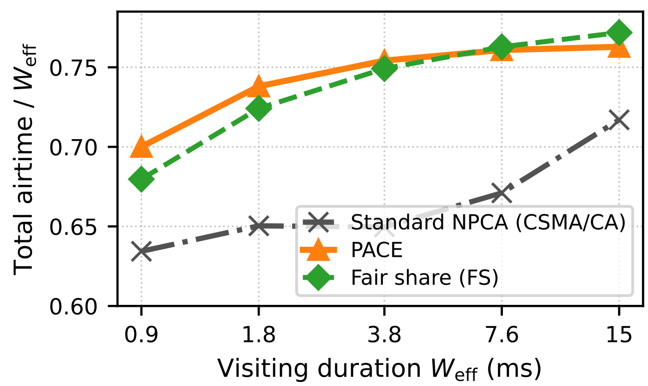


(b) Mandatory RTS/CTS.

Fig. 5: Visitor airtime proportionality $\rho = \big(\text{visitor airtime share}\big)/\big(N_{\text{vis}}/(N_{\text{vis}} + N_{\text{nat}})\big)$ versus the number of visitor STAs N_{vis}, in the same setting as Fig. 4. $\rho = 1$ is the proportional-fair reference; $\rho > 1$ means the visitors occupy more than their population share.



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

Fig. 6: Total useful airtime versus the visiting duration W_{eff} (0.9–15.1 ms), with $N_{\text{vis}} = 20$ visitor STAs and $N_{\text{nat}} = 10$ natives; all other parameters as in Table II.

natives' DCF decrements its backoff only in idle slots and therefore competes sluggishly inside a busy window. Exact proportionality is thus not attainable by any attempt-fair policy in this setting, and the FS curve marks the practically fair operating line.

Against this reference the two schemes separate cleanly. The standard-compliant visitors hold $\rho \approx 1.0$–1.2 throughout. they take at least their population share even while, as Fig. 4a showed, their collision cascades destroy the channel at large N_{vis}, over-proportional and inefficient at once. PACE instead operates *below* unity across the entire range and approaches proportionality from beneath as the visitor demand grows (ρ rising from 0.24 to 0.88 under basic access and from 0.31 to 0.92 under RTS/CTS). This is the fairness counterpart of the flat total-airtime curves of Fig. 4. The conservative start $\tau_0 = 1/W_{\text{eff}}$ concedes the channel to the natives when few visitors are present, and the idle-driven increase claims airtime only as the visitor population, and hence its legitimate share, grows. PACE therefore never extracts its gain from the natives. its advantage over the standard procedure in Section V-A comes from slots that contention had been wasting, not from native airtime.

C. Impact of Visiting Duration

The window an NPCA visit receives is not a design choice but an environmental one. It is the remaining duration of the OBSS transmission that opened the opportunity, and it varies from sub-millisecond frames to TXOP-scale bursts. Fig. 6 therefore sweeps the visiting duration from 0.9 to 15.1 ms (1.6 to 27 mean visitor frames) at $N_{\text{vis}} = 20$ visitors and $N_{\text{nat}} = 10$ natives.

Under basic access (Fig. 6a), the standard visitors hold the channel at 0.33–0.44 across the sweep. However long the visit lasts, $\text{CW}_{\text{min}} = 16$ remains mismatched to the 30-station contention, and the synchronized collision cascades, each charged a full frame length, consume a roughly constant fraction of whatever window is available. PACE sustains 0.63–0.72 across the entire sweep, between $1.6\times$ and $2.0\times$ the standard procedure, and stays at or above FS at every duration, for the reason established in Section V-A. Its conservative, deadline-aware ramp avoids collisions where collisions are expensive.

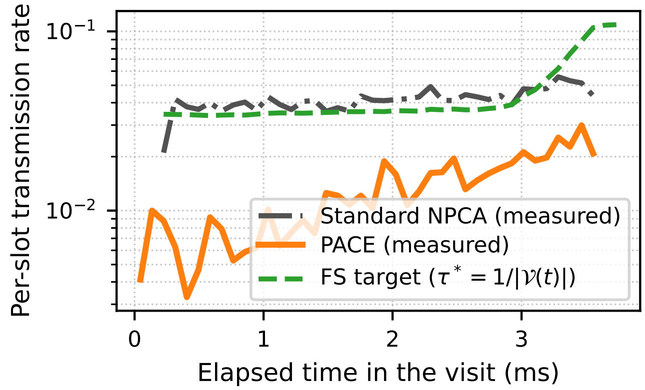
Under mandatory RTS/CTS (Fig. 6b), PACE reaches the FS operating line already at the shortest visit and tracks it over the whole sweep (0.70–0.76), whereas the standard visitors start 10% below at 0.9 ms and need the full 15 ms to close the gap, climbing from 0.63 to 0.72. The margin is smaller than under basic access because the handshake caps the collision cost, but its origin is telling. Within a sub-millisecond visit a backoff process has too few idle slots to drain even one moderate draw, while PACE's per-slot probability is re-steered from the first slot onward. At such durations PACE's own visitors contribute modestly (0.23 of the window, versus 0.39 for the standard visitors), the one-probe budget $\tau_0 = 1/W_{\mathrm{eff}}$ leaves the feedback little room to ramp, but the composition is the fair one. The slots PACE declines to contest carry native traffic, keeping the total above the standard procedure at every duration.

D. Tracking the Time-Varying Fair-Share Target

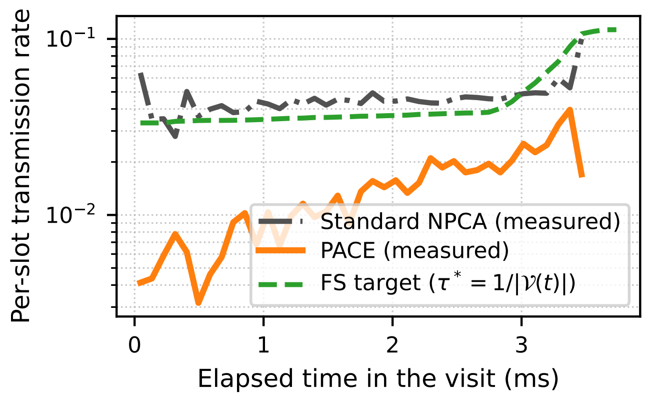
The aggregate results of Sections V-A–V-C follow from how each scheme drives its transmissions inside a visit, which Fig. 7 makes visible. Since the standard visitor holds no per-slot probability, we plot for both schemes the same model-free quantity, the measured fraction of viable visitors transmitting in each slot, against the analytic FS target. The target $\tau^*(t)$ stays near $1/(N_{\mathrm{vis}}+N_{\mathrm{nat}}) = 1/30$ for most of the visit and then rises steeply in the final slots, as successful departures and PPDU-aware self-exclusion drain the viable set. The correct policy must not only adapt but adapt *upward, and increasingly fast, toward the deadline*.

The standard-compliant visitor enters at $1/\mathrm{CW}_{\min} = 1/16$, roughly twice the target, and its measured rate then hovers near the target it cannot see. The aggregate rate, however, conceals the pathology. The countdown is deterministic and freezes during every busy period, so backoff expirations bunch together when the medium clears, and attempts at the same average rate collide far more often than independent ones, 79–81$\%$ of standard-visitor transmissions collide, against 55% for FS at a comparable rate (Table III). The rate is moreover sustained by a rotating minority of lottery winners, while each collision doubles the losers' windows with no in-visit reset, so the per-STA starvation grows even as the aggregate holds.

PACE needs neither the target nor luck. It enters at $\tau_0 = 1/W_{\mathrm{eff}}$, deliberately below the target, and the idle-driven multiplicative increase together with solo-copy propagation raises the measured rate monotonically to the target's level over the course of the visit; being genuinely probabilistic, its attempts stay uncorrelated, and only 40% of them collide. Approaching the target from below is the safe side established in Section IV-A. An under-aggressive probability costs a few idle slots, which the feedback quickly reclaims, whereas over-aggressive or synchronized attempts spend the window on full-frame collisions that no subsequent adaptation can refund. The same qualitative picture holds in both access modes; under RTS/CTS the convergence is tighter because each corrective feedback event costs at most the constant L_{col} rather than a full frame.



(a) Basic access (no RTS/CTS).



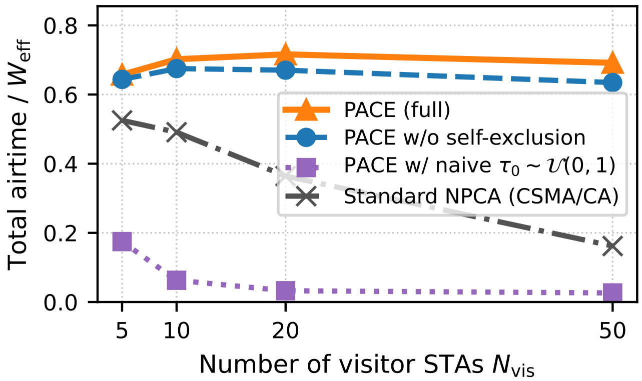
(b) Mandatory RTS/CTS.

Fig. 7: Within-visit transmission dynamics in the reference setting of Fig. 4 ($N_{\mathrm{vis}} = 20$, $N_{\mathrm{nat}} = 10$, $W_{\mathrm{eff}} = 3.78$ ms). The analytic FS target $\tau^*(t) = 1/|\mathcal{V}(t)|$ and the *measured* per-slot transmission rate of viable visitors under PACE and under the standard procedure, averaged over 500 visits.

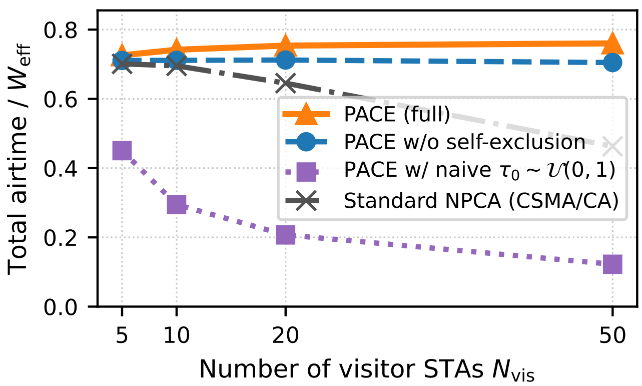
TABLE III: Per-Attempt Outcome of Visitor Transmissions (setting of Fig. 7, steady state)

Access	Scheme	Att./visit	Succ.	Coll.
basic	Standard NPCA	9.3	21%	79%
	PACE	3.8	61%	39%
	FS	7.4	45%	55%
RTS/CTS	Standard NPCA	18.0	19%	81%
	PACE	4.2	60%	40%
	FS	9.4	45%	55%

Table III condenses this into the *contention efficiency* of each scheme. The outcome of every visitor transmission attempt in the setting of Fig. 7. The standard visitor spends 9.3 attempts per visit (and 18.0 under RTS/CTS, where cheap collisions let it retry faster) yet only 19–21$\%$ of them succeed; the rest are synchronized collisions. PACE makes between a



(a) Basic access (no RTS/CTS).



(b) Mandatory RTS/CTS.

Fig. 8: Ablation study in the setting of Fig. 4: total useful airtime for full PACE, PACE without PPDU-aware self-exclusion (Phase (b) removed; unfittable frames contend and are truncated at timer expiry), PACE without the one-probe initialization (knowledge-free random $\tau_0 \sim \mathcal{U}(0,1)$ instead of $1/W_{\text{eff}}$), and the standard-compliant baseline.

quarter and half as many attempts and lands 60–61% of them, a $3\times$ higher per-attempt yield. A high success ratio alone could of course be bought with near-silence; what makes the number meaningful is its pairing with Fig. 4. PACE attains the *highest* total airtime of the three schemes while attempting the least, whereas the standard procedure attempts the most and delivers the least. Beyond airtime, this frugality is itself a system virtue. Every avoided futile attempt is energy not spent and interference not created, which the per-visit attempt counts quantify directly.

E. Ablation Study

PACE adds two mechanisms to the bare MIMD adaptation rule. The one-probe initialization $\tau_0 = 1/W_{\text{eff}}$ of Phase (a) and the PPDU-aware self-exclusion of Phase (b). Fig. 8 removes each in turn in the setting of Fig. 4 to show that neither is a tuning detail. The two mechanisms guard opposite ends of the visit, the initialization protects its beginning from a collision storm, and self-exclusion protects its end from truncation waste.

Replacing the initialization with the knowledge-free choice $\tau_0 \sim \mathcal{U}(0,1)$, the uninformed prior a designer might default to absent any population estimate, is catastrophic. Under basic access the total airtime collapses to 0.18–0.03, an order of magnitude below full PACE and below even the standard baseline at every population. The visit opens in a collision storm whose every collision is charged the longest colliding frame, and the multiplicative decrease, built for gradual convergence, not disaster recovery, cannot pull τ down to the operating point of order 10^{-2} before the window has drained. Protection does not rescue it (0.45–0.12, below the standard baseline throughout), because the failure is not only the storm's cost but a consensus breakdown. The visitors drawn near $\tau \approx 1$ transmit in almost every slot, so solo successes , the events that propagate a common operating point through the copy rule of (11), almost never occur, and the ensemble has no mechanism left to agree on a lower τ. A fixed naive $\tau_0 = 0.5$ fails in the same way under basic access and only somewhat less severely under protection, where its homogeneous start at least descends in lockstep. The comparison also exposes a fairness failure. at $N_{\text{vis}} = 5$ under RTS/CTS the aggressive visitors seize 0.40 of the channel while the ten natives retain 0.05. A probabilistic access scheme without a deadline-scaled start is thus *worse* than the backoff procedure it replaces, the conservative initialization is a precondition for PACE, not an optimization.

Removing self-exclusion costs less but systematically, and the loss grows with the visitor population. Total airtime drops by 2% at $N_{\text{vis}} = 5$ and by 8% at $N_{\text{vis}} = 50$ under basic access (0.635 versus 0.692), and by 7% at $N_{\text{vis}} = 50$ under RTS/CTS (0.705 versus 0.760). Without Phase (b), a visitor whose frame no longer fits keeps contending, inflating the collision rate seen by the STAs that could still succeed, and, when it does win a slot, starts a transmission that the NPCA timer cuts off, forfeiting the window tail. Self-exclusion converts both losses into usable airtime for the remaining viable STAs. That the ablated variant still far exceeds the standard baseline confirms the MIMD adaptation as the dominant mechanism; self-exclusion is the complementary guard that keeps its gains from eroding at the deadline.

VI. Conclusion

This paper presented PACE, a probabilistic adaptive contention control scheme for non-primary channel access in IEEE 802.11bn. The bounded NPCA visit inverts the premise of BEB. As the window drains and the viable population shrinks, the fair operating point $\tau^*(t) = 1/|\mathcal{V}(t)|$ rises, yet BEB retreats on every collision, idles behind frozen counters, and bunches its expirations into synchronized collisions. PACE replaces the backoff countdown with a per-slot transmission probability governed by three purely local rules. The one-probe initialization $\tau_0 = 1/W_{\text{eff}}$, PPDU-aware self-exclusion at the deadline, and multiplicative feedback adaptation with solo-copy consensus. No central coordination, message exchange beyond the advertised probability, or knowledge of the contending population is required.

Evaluated in IEEE 802.11 standard units against a standard-compliant CSMA/CA baseline and a genie-aided fair-share reference, PACE kept the total useful airtime of the shared channel within 2% of the fair-share reference across 5–50 visitors under RTS/CTS-protected access, up to 63% above the standard procedure, and under basic access held the channel where the standard procedure collapsed, reaching a $4.5\times$ advantage. The gain is not extracted from the incumbents. PACE's visitors stay at or below their population-proportional airtime share, whereas the standard visitors over-claim theirs even while destroying the channel. The advantage persists across visit durations from one to fifteen milliseconds, growing with the contending population, and PACE achieves it while transmitting the least. a fraction of the baseline's attempts with a $3\times$ higher per-attempt success ratio, a direct saving in energy and interference. The ablation study confirmed that these properties rest on both added mechanisms. Without the deadline-scaled initialization the probabilistic access collapses below the baseline it replaces, and without self-exclusion its gains erode by up to 8% at the deadline.

Several directions remain open. The symmetric MIMD factors balance event counts rather than airtime costs, leaving the equilibrium slightly over-aggressive when collisions are charged at full frame length; a cost-aware decrease factor weighted by the observed collision occupancy would move the equilibrium toward the airtime optimum under basic access. The single-collision-domain model bounds the interaction with native STAs; partial-overlap topologies with spatial reuse merit a dedicated study. Relaxing the one-frame-per-visit abstraction toward TXOP aggregation, and warm-starting τ across repeated visits to shorten the ramp-up in sub-millisecond windows, with a correction for the inflation such carried state accumulates, are natural extensions of the present design.

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